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MILITARY SPECIFICATION

# LUBRICATING UNIT, POWER-OPERATED, ELECTRIC, PORTABLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers a portable, electrically powered lubricating unit for dispensing fluid or viscous lubricants for airframes and chassis.

## 2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

| | |
|---|---|
| W-C-596/13 | - Connector, Plug, Electrical, General Purpose, Hospital Grade Grounding, Two Pole, Three Wire, 15 Amperes, 125 Volt, 50/60 Hertz. |
| QQ-S-781 | - Strapping, Steel, and Seals. |
| PPP-B-621 | - Boxes, Wood, Nailed and Lock-Corner. |
| PPP-P-40 | - Packaging and Packing of Hand Tools. |
| PPP-T-60 | - Tape: Packaging, Waterproof. |

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research and Development Center, ATTN: STRBE-DS, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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| MIL-P-116 | - Preservation-Packaging, Methods of. |
| MIL-B-121 | - Barrier Material, Greaseproofed, Waterproofed, Flexible. |
| MIL-V-173 | - Varnish, Moisture-and-Fungus-Resistant, (for the Treatment of Communications, Electronic, and Associated Equipment). |
| MIL-P-514 | - Plates, Identification, Instruction and Marking, Blank. |
| MIL-T-704 | - Treatment, and Painting of Materiel. |
| MIL-H-3868 | - Hose Assemblies, Grease Gun, High and Low Pressure. |
| MIL-L-4387 | - Lubrication Dispensing Equipment Accessories. |
| MIL-E-16298 | - Electric Machines Having Rotating Parts and Associated Repair Parts, Packaging of. |
| MIL-G-20241 | - Gasket Material, Wool Felt, Impregnated, Adhesive, Pressure-Sensitive. |
| MIL-G-23827 | - Grease, Aircraft and Instrument, Gear and Actuator Screw. |
| MIL-S-81733 | - Sealing and Coating Compound, Corrosion Inhibitive. |

STANDARDS

FEDERAL

| | |
|---|---|
| FED-STD-H28 | - Screw Threads Standards for Federal Services. |

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|---|---|
| MIL-STD-105 | - Sampling Procedures and Tables for Inspection by Attributes. |
| MIL-STD-129 | - Marking for Shipment and Storage. |
| MIL-STD-130 | - Identification Marking of U.S. Military Property. |
| MIL-STD-461 | - Electromagnetic Emission & Susceptibility Requirements for Control of Electromagnetic Interference. |
| MIL-STD-462 | - Electromagnetic Interference Characteristics, Measurement of. |
| MIL-STD-889 | - Dissimilar Metals. |
| MIL-STD-1186 | - Cushioning, Anchoring, Bracing, Blocking, and Waterproofing, with Appropriate Test Methods. |
| MIL-STD-1472 | - Human Engineering Design Criteria for Military Systems, Equipment & Facilities. |



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(Copies of specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN GEAR MANUFACTURER'S ASSOCIATION (AGMA)

AGMA 390.03 - Gear Handbook, Volume 1.

(Application for copies should be addressed to the American Gear Manufacturer's Association, 1330 Massachusetts Avenue, NW, Washington, DC 20005.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/AGMA 201.02 - Tooth Proportions For Coarse-Pitch Involute Spur Gears.
ANSI/AWS D1.1 - Structural Welding Code, Section 5, Qualification.

(Application for copies should be addressed to the American National Standards Institute, 1439 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practices for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

SOUTH COAST AIR QUALITY MANAGEMENT DISTRICT

Rule 442 - Usage of Solvents.

(Application for copies should be addressed to the South Coast Air Quality Management District, 9150 Flair Drive, El Monte, CA 91731.)

NATIONAL ELECTRICAL MANUFACTURER'S ASSOCIATION (NEMA)

MG-1 - Motors and Generators.



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(Application for copies should be addressed to the National Electrical Manufacturer's Association, 155 East 44th Street, New York, NY 10017.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA No. 70 - National Electrical Code.

(Application for copies should be addressed to the National Fire Protection Association, 470 Atlantic Avenue, Boston, MA 02110.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The lubricating unit, power-operated, electrical, portable (hereinafter called "lubricator") shall consist of a chassis mounted on two wheels and one swivel-type caster, a handle, lubricant container, automatic mechanical feed device, pressure pump, discharge hose, control valve, swivels, pressure switch, safety valve, racks, electric motor, and power supply cord.

3.2 First article. When specified, a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.3 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Material deterioration and control. The lubricator shall be fabricated from compatible materials, inherently corrosion resistant or treated to be resistant against the various forms of corrosion and deterioration that may be encountered in any of the applicable storage or operating environments to which the item may be exposed.

3.3.1.1 Dissimilar metals. Dissimilar metals as defined in MIL-STD-889, shall be electrically insulated from one another to minimize or prevent galvanic corrosion. Insulation may be provided by a barrier such as a corrosion inhibiting sealant conforming to MIL-S-81733 or chromate tape conforming to MIL-G-20241.

3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components and sub-components, and shall make information available, upon request to the contracting officer or designated representative.



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3.3.2 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the lubricator may be newly fabricated from recovered materials to the maximum extent practicable, provided the lubricator produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the lubricator.

3.4 Environmental conditions.

3.4.1 Storage temperatures. The lubricator shall be capable of withstanding storage without damage in a temperature of -30° F to +160° F. The lubricator shall be tested for performance in accordance with 4.5.2.7 and 4.5.2.8.

3.4.2 Operating temperature. The lubricator shall operate effectively while being subjected to an operating temperature of 0° F to +110° F when tested in accordance with 4.5.2.5.1 and 4.5.2.5.2.

3.5 Safety. Each lubricator shall be effectively guarded and grounded to protect all persons and objects from electrical shock hazard. All belts, pulleys, and other reciprocating, rotating, or moving parts of equipment shall be guarded when such parts are exposed to contact by personnel or otherwise create a hazard. Bonding as specified by NFPA No. 70, Article 501.16, shall apply when operating equipment in class 1, division 1 and 2 locations. External surfaces shall be free of burrs, sharp edges, and corners except where sharp edges or corners are required or where they are not detrimental to safety.

3.6 Human factors engineering. The lubricator shall conform to accepted human factors engineering design criteria as described in MIL-STD-1472. Special design emphasis shall be given, but not limited to paragraphs 4 (General requirements), 5.2 (Visual displays), 5.4 (Controls), 5.5 (Labeling), 5.9 (Design for maintainability), and 5.13 (Hazards and safety) of MIL-STD-1472 as applicable. The lubricator shall be operable and maintainable by the 5th through 95th percentile soldiers who are dressed appropriately for the environments of interest.

3.7 Design. The lubricator shall be portable, electrically powered, and capable of delivering lubricant at an operating pressure of 6000 pounds per square inch gage (psig).

3.7.1 Threads. All threads shall be in accordance with FED-STD-H28.

3.7.2 Fasteners. All screws, bolts, and similar parts shall be installed with a means for preventing loss of tightness and adjustment. All such parts subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently fastened.

3.7.3 Gears. All gears and pinions shall be in accordance with AGMA 390.03. Spur gear tooth form shall be in accordance with ANSI/AGMA 201.02.



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3.7.4 Lubrication. All bearings and surfaces requiring lubrication shall be provided with a means for lubricating.

3.7.5 Packing and gaskets. Packing and gaskets used in the lubricator shall be of corrosion- and grease-resistant material capable of withstanding all the pressures and operating conditions specified herein.

3.7.6 Electrical connections. Electrical connections shall be mechanically secured and soldered or brazed, except connections on screw-type terminals which shall be mechanically secured.

3.8 Components.

3.8.1 Portable chassis. The lubricator shall be mounted on an integral chassis. The chassis shall be mounted on two wheels and one swivel-type caster. The two wheels shall be not less than 8 inches in diameter, having semi-pneumatic rubber tires with not less than 2-inch tread width and equipped with antifriction-type bearings. The front caster shall be not less than 4 inches in diameter, having a semi-pneumatic rubber tire with not less than 1-1/2-inch tread width and equipped with antifriction-type bearings or a solid rubber tire with antifriction-type bearings. The chassis shall provide for mounting of the grease container, electric motor, pump, hose and cord racks, control valve, pressure switch, and handle.

3.8.2 Handle. The handle shall be of permanent or movable type. If a movable handle is furnished, it shall be provided with means for locking in the vertical position. The handle shall be attached to the lubricator and shall be not less than 36 inches nor more than 42 inches from the floor level when in the vertical position.

3.8.3 Lubricant container. The lubricant container shall be sheet steel not less than 18 gage (0.0478-inch nominal thickness). The container shall have a capacity of 30 to 40 pounds, ±5 pounds, of lubricant conforming to MIL-G-23827.

3.8.4 Automatic mechanical feeding device. The automatic mechanical feeding device shall wipe the lubricant from the sides of the container and force the lubricant from the sides of the container to the pressure pump below.

3.8.5 Pressure pump. The pressure pump shall be capable of providing a grease discharge pressure of not less than 6000 psig without evidence of external leakage, distortion, or damage. The pressure shall not drop more than 1000 psig in 5 minutes. The pump shall be self-priming and shall be provided with an air bleeder valve. The pump outlet shall be internally threaded 1/4-18 NPTF.

3.8.6 Discharge hose. The discharge hose shall conform to MIL-H-3868, type I, style 2, grade A, and shall be not less than 15 feet long with 1/4-inch inside diameter. The hose end fittings shall be rigid and mate with the swivels as specified in 3.8.8.



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3.8.7 Control valve. The control valve shall conform to MIL-L-4387, type VI, class 1, style A, except that the grease inlet shall be internally threaded 1/4-18 NPTF.

3.8.8 Swivels. The two swivel fittings shall conform to MIL-L-4387. One swivel shall conform to type VII, class 1, style A, except that the size shall be 1/4-18 NPTF external to 1/4-18 NPTF external, and shall be provided between the pump outlet and discharge hose. The other swivel shall conform to type VII, class 3, except that the size shall be 1/4-18 NPTF external to 1/8-27 NPTF external, and shall be provided between the discharge hose and control valve.

3.8.9 Pressure switch. The automatic pressure switch shall control the discharge pressure of the lubricator between 4000 psig, +100 or -400 psig, and 6000 psig, +400 or -100 psig. The switch shall stop the motor when a pressure of 6000 psig, +400 or -100 psig, has been attained, and shall start the motor when the pressure has dropped to 4000 psig, +100 or -400 psig.

3.8.10 Safety valve. The safety valve shall bypass the grease back into the container to safeguard the lubricator when a pressure not to exceed 6600 psig has been reached. In addition, the safety valve shall activate a manual reset-type circuit breaker provided as a back-up should the main pressure switch specified in 3.8.9 fail.

3.8.11 Racks. Racks shall be provided for the coiling and transporting of the discharge hose and power supply cord. Racks may be a vertical handle type, or provisions shall be made for coiling hose and electrical leads around the grease container.

3.8.12 Electrical.

3.8.12.1 Motor. Unless otherwise specified (see 6.2), the motor shall conform to NEMA MG-1, alternating current (ac), 120 volts, not less than 3/8 horsepower (hp), 60 hertz (Hz), single phase, continuous duty, and explosion-proof.

3.8.12.2 Electrical components. Electrical components of the lubricator, including relays, switchs, wiring, and their inclosures, shall conform to and shall be located in accordance with the applicable NEMA electrical standards, except as specified herein. No component or part shall be used in any application where the requirements exceed the manufacturer's rating for the part. Electrical components shall be explosionproof as specified by NFPA No. 70, Article 501-4, for class 1, division 1 locations.

3.8.12.3 Power supply cord. A 3-conductor, type S, heavy-duty supply cord not less than 25 feet long as specified in NFPA No. 70, Article 501-1 shall be furnished with each lubricator. The free end of the cord shall be provided with a connector plug conforming to W-C-596/13.



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3.8.12.4 Dielectric strength. The electrical circuits of the lubricator shall be capable of withstanding, without damage, the application of 1000 volts root mean square (rms), 60 Hz, ac, plus twice the normal operating voltage for not less than 1 minute, when applied between the frame and current-carrying parts.

3.9 Performance. The lubricator shall be capable of discharging grease conforming to MIL-G-23827 at the pressure, temperature, and volume specified in table I, with unrestricted flow (free flow) and with 1000 psig back pressure while operating at a discharge pressure of 4000 psig, +100 or -400 psig, to 6000 psig, +400 psig or -100 psig, at the control valve.

TABLE I. Discharge capacity requirements.

| Test Function | Minimum Requirement | Reference Paragraph |
|---|---|---|
| Minimum rate of discharge at +70° F, free flow. | 14 oz. per min. | 4.5.2.5.1 |
| Minimum rate of discharge at +70° F against 1000 psig back pressure. | 12 oz. per min. | 4.5.2.5.1 |
| Minimum rate of discharge at 0° F, free flow. | 12 oz. per min. | 4.5.2.5.2 |
| Minimum rate of discharge at 0° F, against 1000 psig back pressure. | 8 oz. per min. | 4.5.2.5.2 |

3.10 Discharge pressure. The lubricator shall be capable of producing a discharge pressure of 6000 psig, +400 or -400 psig, of the discharge nozzle. The lubricator discharge pressure shall be controlled between 4000 psig, +100 or -400 psig, and 6000 psig, +400 or -100 psig, by means of an automatic pressure switch.

3.11 Electromagnetic interference. Unless otherwise specified (see 6.2), electromagnetic interference emission characteristics of the lubricator shall conform to MIL-STD-461, class IIB.

3.12 Fungus and moisture resistance. When specified (see 6.2), the electrical circuitry, including all components and connections, except as specified below, shall be protected from the effects of fungus growth and moisture by an overall treatment with a varnish conforming to MIL-V-173, composition I or II, as applicable (see 3.12.1), with 1 percent copper 8-quinolinolate (by weight) based on the nonvolatile content of the varnish.

a. Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.



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b. Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a dry-film thickness of not less than 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.12.1 Composition. Composition II shall be used only in the case when local air pollution regulations governing the application of varnish precludes the use of composition I. When composition II is used, the contractor shall provide evidence to the Government that the use of composition II is required, and shall certify that the composition II material complies with Rule 442 South Coast Air Quality Management District, El Monte, CA.

3.13 Identification marking. The lubricator shall be identified in accordance with MIL-STD-130. The marking shall be applied to the lubricator on plates conforming to MIL-P-514, type I, style 3, composition C of type I, grade A, class 1 material. Each plate shall be attached by screws, bolts, or rivets in a location where the plate will be both visible and legible.

3.13.1 Warning markings. A warning plate shall be conspicuously posted on the lubricant container with the inscription, "WARNING: NEVER DIRECT NOZZLE TOWARD PERSONNEL". The warning plate shall be legible under average light conditions from a distance of 15 feet. The warning plate shall conform to type III of MIL-P-514. Unless otherwise specified (see 6.2), plate material and type of protective coating shall be at the option of the contractor.

3.14 Treatment and painting. The portions of the lubricator normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, type A, color as specified (see 6.2). The interior of gear cases or similar compartments and cast-in reservoirs shall be protected by a sealer that will withstand immersion in lubricating oil and hydraulic fluids at working temperatures and atmospheric conditions. Unless otherwise specified (see 6.2), the sealer used shall be at the option of the contractor.

3.15 Workmanship.

3.15.1 Castings and forgings. All parts, components, and assemblies of the lubricator which include castings and forgings shall be clean of harmful extraneous material such as sand, dirt, sprues, scale, and flux. Rework shall be limited to procedures which do not reduce mechanical properties or affect function.

3.15.2 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, mill scale that can be removed by chipping and wire brushing, and other foreign matter. Parent materials, weld filler metals, and fabrication techniques shall be as required to enable the lubricator to conform



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to the examination and test requirements specified in section 4. Parts to be joined by fillet welds shall be brought into as close contact as possible and in no event shall be separated by more than 3/16 inch unless appropriate bridging techniques are used. Unless otherwise specified (see 6.2), the welding process used in fabrication of the lubricator shall be at the option of the contractor.

3.15.2.1 Welding of aluminum. The aluminum surfces of parts to be welded shall be free from scale, paint, grease, oxide film, and other foreign matter. The welding shall be done by the inert gas, shielded-arc method. Welds may be ground, filed, buffed, or chipped but shall not be hammered, and shall be free from discontinuities such as base metal cracks adjacent to or behind welds, cracks in weld metal, undercutting of base metal, lack of fusion with parent metal or between parts, lack of penetration, lack of prescribed fit, or spatter.

3.15.2.2 Welders. Before assigning any welder to manual welding work covered by this specification, the contractor shall obtain certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the type of welding operations to be performed and that such qualification is effective as defined by the particular code:

ANSI/AWS D1.1 - Structural Welding Code - Steel.
ANSI/AWS D1.2 - Structural Welding Code - Aluminum.
Welding Qualifications of the ASME, Section IX.

Contractors who make only horizontal welds need not qualify welders for "all position welding". The contractor shall be responsible for determining that automatic welding equipment operators are qualified in accordance with ANSI/AWS or ANSI/ASME codes. The certification shall be made available for review by the contracting officer or the contracting officer's representative.

3.15.2.3 Welding practices.

a. Tack welds shall be subject to the same quality requirements as final welds.
b. Work shall be positioned for flat welding whenever practicable.
c. Procedures and sequences shall be such that distortion and shrinkage will be held to a minimum. When straightening is required, caution shall be exercised to insure that the straightening process does not weaken the part and lubricator.

3.15.2.4 Weld quality.

a. The contractor shall be responsible for the inspections needed to insure that, when the welded parts are assembled together to make the lubricator, the lubricator shall conform to the inspection requirements specified in section 4.
b. All weldments shall be free of slag, flux, weld spatter, cracks, incomplete penetration and other discontinuities which are not permitted in ANSI/AWS D1.1 Structural Welding Code.



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c. Undercut in weldments shall be held to a minimum and shall be not more than 1/32 inch deep.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. The requirements specified herein are classified as follows:

a. First article lubricator inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 First article lubricator inspection.

4.3.1 Examination. The first article lubricator shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The first article lubricator shall be tested as specified in 4.5.2 through 4.5.2.9. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined for the major and minor defects specified in 4.5.1. AQL shall be 2.5 percent defective for major defects and 4.0 percent defective for minor defects.

4.4.3 Tests. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.5.2 through 4.5.2.6. AQL shall be 2.5 percent defective.

4.5 Inspection procedure.

4.5.1 Examination. The lubricator shall be examined as specified herein for the following defects:



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Major

101. Material not as specified.

a. Materials are not corrosion resistant or treated to be made corrosion resistant for the applicable storage or operating environment.
b. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.
c. Contractor does not have documentation available for identification of material, material finishes or treatments.

102. Safety requirement not as specified.
103. Design not as specified.
104. Components missing or not as specified.
105. Fungus and moisture resistance not as specified (when required).
106. Workmanship not as specified.
107. Human factors engineering requirements not as specified.
108. Portable chassis not as specified.
109. Handle not as specified.
110. Lubricant container not as specified.
111. Automatic mechanical feeding device not as specified.
112. Pressure pump not as specified.
113. Discharge hose not as specified.
114. Control valve not as specified.
115. Swivels not as specified.
116. Pressure switch not as specified.
117. Safety valve not as specified.
118. Racks not as specified.
119. Electric motor not as specified.
120. Electrical components not as specified.
121. Power supply cord not as specified.
122. Dielectric strength not as specified.
123. Threads not as specified.
124. Fasteners not as specified.
125. Gears not as specified.
126. Lubrication not as specified.
127. Packing and gaskets not as specified.
128. Electrical connections not as specified.
129. Castings and forgings not as specified.
130. Welding not as specified.
131. Welding of aluminum not as specified.
132. Welder certificate not furnished.
133. Welding practices not as specified.
134. Weld quality not as specified.
135. Composition not as specified.
136. Discharge pressure not as specified.

Minor

201. Identification and warning marking missing, incomplete, or illegible.
202. Treatment and painting not as specified.



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4.5.2 Tests.

4.5.2.1 Static pressure. The hydraulic coupler shall be attached to a hydraulic fitting connected to a pressure gage with not less than 0 to 10,000 psig dial reading. With the outlet closed, the lubricator shall be operated until a pressure of 6000 psig has been registered on the gage dial and then turned off. The coupler, control valve, and hose assembly shall be inspected. Nonconformance to 3.8.5 shall constitute failure of this test.

4.5.2.2 Pressure control switch. After the lubricator has been tested for discharge pressure as specified in 4.5.2.6, the pressure switch shall cut off the motor when the gage indicates 6000 psig, +400 or -100 psig. The lubricator shall be operated until the pressure has dropped to 4000 psig, +100 or -400 psig, at which point the motor shall start operating again. Nonconformance to 3.8.9 shall constitute failure of this test.

4.5.2.3 Safety valve. The hydraulic coupler shall be attached to a lubricator fitting connected to a pressure gage with not less than 0 to 10,000 psig dial reading. The lubricator shall be operated until it has attained the 6000 psig, +400 or -100 psig, discharge pressure with the control valve closed. With the pressure switch contacts bridged, the pump shall continue to operate until the safety valve is activated. The pressure reading on the gage shall not exceed 6600 psig. Nonconformance to 3.8.10 shall constitute failure of this test.

4.5.2.4 Dielectric strength. The electrical circuit of the lubricator shall be subjected to 1000 volts rms, plus twice the normal operating voltage between the frame and current-carrying parts for not less than 1 minute. Nonconformance to 3.8.12.4 shall constitute failure of this test.

4.5.2.5 Performance tests.

4.5.2.5.1 Discharge capacity at +70° F. The lubricator and grease shall be stabilized in an ambient temperature of +70° F, ±5° F, during this test. The lubricator shall be filled with grease conforming to MIL-G-23827 and the grease shall be discharged into a container of known weight and volume during the test. The control valve shall be adjusted for continuous flow. The hydraulic coupler shall be attached to a hydraulic fitting and the lubricator operated free flow for not less than 5 minutes. During this 5-minute period, the lubricator shall have discharged not less than 14 ounces per minute. The grease dispensed shall be weighed to determine conformance to 3.9. The lubricator shall also be operated an additional 5 minutes under the above conditions, except that it shall be required to operate against a back pressure of 1000 psig and discharge not less than 12 ounces per minute. For this phase of the test, the coupler shall be attached to a lubricator fitting connected to a pressure gage with not less than a 0 to 10,000 psig dial reading and a pressure valve. The pressure valve shall be adjusted to a back pressure of 1000 psig. The grease dispensed during this 5-minute period shall be weighed. Nonconformance to 3.9 shall constitute failure of this test.



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4.5.2.5.2 Discharge capacity at 0° F. The same procedure shall be employed in performing this test as in 4.5.2.5.1, except that the lubricator and grease shall be stabilized in an ambient temperature of 0° F, ±5° F, and shall dispense grease at the rate of not less than 12 ounces per minute free flow for 5 minutes and not less than 8 ounces per minute against a back pressure of 1000 psig for 5 minutes. Nonconformance to 3.9 shall constitute failure of this test.

4.5.2.6 Discharge pressure. The lubricator shall be filled with grease conforming to MIL-G-23827. The lubricator shall be operated with the control valve in the open position. The test shall require the hydraulic coupler of the control valve to be attached to a lubricator fitting connected to a pressure gage with not less than 0 to 10,000 psig dial reading. The lubricator shall build up pressure to 6000 psig, +400 or -100 psig. Nonconformance to 3.10 shall constitute failure of this test.

4.5.2.7 Cold storage. The lubricator shall be conditioned at a temperature of -30° F, ±5° F, for a period of 72 hours. The lubricator shall be inspected for evidence of damage and tested for performance in accordance with 4.5.2.5.2.

4.5.2.8 Hot storage. The lubricator shall be conditioned at a temperature of +160° F, ±5° F, for a period of 8 hours. The lubricator shall be inspected for evidence of damage and tested for performance in accordance with 4.5.2.5.1.

4.5.2.9 Electromagnetic interference. The first article lubricator shall be tested in accordance with MIL-STD-462 to determine conformance to 3.11. During this test, the unit shall be operated so as to exercise the pressure switch(es). The contractor shall furnish the contracting officer a report of tests required by MIL-STD-461. Disapproval of the report shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

| | | |
|---|---|---|
| 137. | Materials, methods, or containers not as specified. | Each incorrect material, method, or container shall constitute one defect. |
| 138. | Disassemly not as specified for level A. | |
| 139. | Hose not coiled and secured as specified for level A. | |
| 140. | Power cord not coiled and secured as specified for level A. | |
| 141. | Electric motor not preserved in accordance with referenced document for level A. | |



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142. Maintenance tools not preserved in accordance with referenced document for level A.
143. Straping not as specified for level A.
144. Blocking, bracing, anchoring, and cushioning not as specified for level A or level B.
145. Marking illegible, incorrect, or incomplete.
146. Repair parts not preserved in accordance with the applicable criteria of MIL-P-116.
147. Technical publications not preserved in accordance with MIL-P-116, method IC-1.
148. Commercial preservation and packing not in accordance with ASTM D 3951.
149. Marking illegible, incorrect, incomplete or otherwise not as specified in referenced document.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or commercial, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Disassembly. The wheels, caster, handle, and any other parts that can be removed to effect a reduction in cube or which are vulnerable to damage and which can be reinstalled without special tools shall be disassembled from the basic unit. Removed nuts, bolts, washers, and screws shall be reinstalled in one of the mating parts and secured to prevent their loss, or they shall be preserved in accordance with MIL-P-116, method IC-1.

5.1.1.2 Preservatives. Preservatives specified herein by "P" numbers shall conform to the applicable specifications listed in and shall be applied in accordance with MIL-P-116.

5.1.1.3 Unpainted metal surfaces. Unpainted exterior metal surfaces not specified herein requiring the application of a contact preservative in accordance with MIL-P-116 shall be coated with type P-1 preservative. Exterior machined surfaces shall be coated with type P-6 or P-11 preservative, wrapped with barrier material conforming to MIL-B-121, type I, grade A, class 2, and secured with tape conforming to PPP-T-60, type IV. Internal metal surfaces requiring a preservative shall be coated with type P-10, type I, grade 10 preservative, and allowed to drain.

5.1.1.4 Hose. The hose shall be coiled on the rack provided. The open end of the hose shall be closed with tape meeting the requirements of PPP-T-60, type IV. The coiled hose shall be secured to the rack to prevent uncoiling.

5.1.1.5 Power cord. The cord shall be coiled on the rack provided and secured.



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5.1.1.6 Electric motor. The electric motor shall be preserved in accordance with MIL-E-16298, level A, by the alternate preservation method as specified for equipment assembled to machines.

5.1.1.7 Repair parts. Such repair parts as are provided shall be preserved in accordance with the applicable criteria of MIL-P-116.

5.1.1.8 Maintenance tools. The maintenance tools provided shall be preserved in accordance with the level A requirements of PPP-P-40.

5.1.1.9 Technical publications. The technical publications provided shall be preserved in accordance with MIL-P-116, method IC-1.

5.1.2 Commercial. Commerical preservation shall be in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, level B, or commercial, as specified (see 6.2).

5.2.1 Level A. Each lubricator, with removed parts and accessories, shall be packed in a close-fitting box conforming to PPP-B-621, class 2, type 3 load. When the wheels and caster are not removed from the lubricator, lumber blocking shall be used to remove all weight from the tires. Anchoring, blocking, bracing, and cushioning the contents of each shipping container shall be in accordance with MIL-STD-1186. Strapping shall conform to QQ-S-781, class 1, type I or IV, size as applicable. Unless otherwise specified (see 6.2), strapping shall be finish B. When specified (see 6.2), strapping shall be finish A.

5.2.2 Level B. Each lubricator, with removed parts and accessories, shall be packed as specified in 5.2.1 for level A, except that the boxes shall be class 1 and strapping shall be either finish A or finish B.

5.2.3 Commercial. Commercial packing shall be in accordance with ASTM D 3951.

5.3 Marking.

5.3.1 Military packaging. Marking shall be in accordance with MIL-STD-129.

5.3.2 Commercial packaging. Marking shall be in accordance with ASTM D 3951. In addition, the shipping weight and cube shall also be marked on the shipping container.

6. NOTES

6.1 Intended use. The lubricator covered by this specification is intended to be used for the lubrication of automotive vehicles and aircraft.

6.2 Ordering data.



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6.2.1 Acquisition documents. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Time frame required for submission of first article lubricator, and number of lubricators required (see 3.2).
c. Motor, if other than specified (see 3.8.12.1).
d. Electromagnetic interference emission characteristics, if other than specified (see 3.11).
e. When fungus and moisture resistance is required (see 3.12).
f. When the warning plate material shall not be at the option of the contractor (see 3.13.1).
g. Color required (see 3.14).
h. When sealer is not at the option of the contractor (see 3.14).
i. When the welding process is not at the option of the contractor (see 3.15.2.1).
j. Degree of preservation and packing required (see 5.1 and 5.2).
k. When strapping shall be other than as specified (see 5.2.1).

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, contract data requirements list (CDRL), the data requirements identified below shall be developed as specified by an approved data item description (DD Form 1644) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9 (n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

| Paragraph No. | Data requirements | Applicable DID No. | Options |
|---|---|---|---|
| 3.15.2.2 | Certification, welder qualification | DI-H-24029 | |

6.3 First article. When a first article inspection is required, the item should be a preproduction item. The first article should consist of one or more acquisition documents regarding arrangements for examinations, tests and approval of the first article test results and disposition of the document's first article.

6.4 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor's maintenance and operation manual to be furnished with each lubricator.

6.5 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions regarding shipment of lubricators.



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6.6 Classification change. The type I lubricators have been discontinued since they are no longer required.

Custodians:
Army - ME
Navy - AS
Air Force - 99

Review activity:
DLA-CS

User activity:
Navy - MC

Preparing activity:
Army - ME

Project 4930-0303



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions – Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
| MIL-L-7312E | Lubricating Unit, Power-Operated, Electric, Portable |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR |
| | ☐ USER |
| b. ADDRESS *(Street, City, State, ZIP Code)* | ☐ MANUFACTURER |
| | ☐ OTHER *(Specify)*: _____ |

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

| 7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional | b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
|---|---|
| c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional | 8. DATE OF SUBMISSION *(YYMMDD)* |

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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